

PostgreSQL Enterprise Naming Standard

Purpose

This standard provides a consistent PostgreSQL naming convention focused on readability, maintainability, relational modeling principles, long-term scalability, and reporting friendliness.

1. General Rules

Use lowercase snake_case for all database objects.

Why: Avoids quoted identifiers and aligns with PostgreSQL practices.

2. Schemas

Examples: sales, finance, master_data, audit, integration.

Why: Schemas provide domain separation and are preferred over table prefixes.

3. Tables

Use singular nouns: customer, order, product, invoice, order_line.

Why: Each row represents a single business entity.

4. Primary Keys

Use entity-specific names: customer_id, order_id, product_id.

Why: Self-documenting and easier to understand in joins.

5. Foreign Keys

Use the same name as the referenced primary key.

Why: Relationship intent is immediately clear.

6. Business Columns

Examples: customer_name, email_address, order_date, invoice_number.

Why: Names should describe the business data.

7. Boolean Columns

Use prefixes is_, has_, can_.

Examples: is_active, has_access, can_approve.

8. Date and Time Columns

Use created_at, updated_at, deleted_at, processed_at.

Use TIMESTAMPTZ.

9. Constraints

PK: pk_customer

FK: fk_order_customer

UQ: uq_customer_email

CHK: chk_order_total_positive

Why: Easier troubleshooting and governance.

10. Indexes

Format: idx_<table>_<column>

Example: idx_customer_email.

11. Views

Prefix: vw_

Example: vw_active_customer.

12. Materialized Views

Prefix: mv_

Example: mv_monthly_sales.

13. Functions

Prefix: fn_

Example: fn_calculate_tax.

14. Triggers

Prefix: trg_

Example: trg_customer_audit.

15. Audit Tables

Prefix: audit_

Example: audit_customer.

16. DDL Standards

Use schema-qualified names, NOT NULL, explicit constraints, TIMESTAMPTZ (includes the timezone), identity columns and NUMERIC(p,s) for financial values.

Naming Examples

sales.customer

customer_id

customer_name

email_address

is_active

created_at

sales.order

order_id

customer_id

order_date

total_amount

sales.order_line

order_line_id

order_id

product_id

quantity
unit_price

DDL Example

```
CREATE TABLE sales.customer (  
  customer_id BIGINT GENERATED ALWAYS AS IDENTITY,  
  customer_name TEXT NOT NULL,  
  email_address TEXT NOT NULL,  
  is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  CONSTRAINT pk_customer PRIMARY KEY (customer_id),  
  CONSTRAINT uq_customer_email UNIQUE (email_address)  
);
```